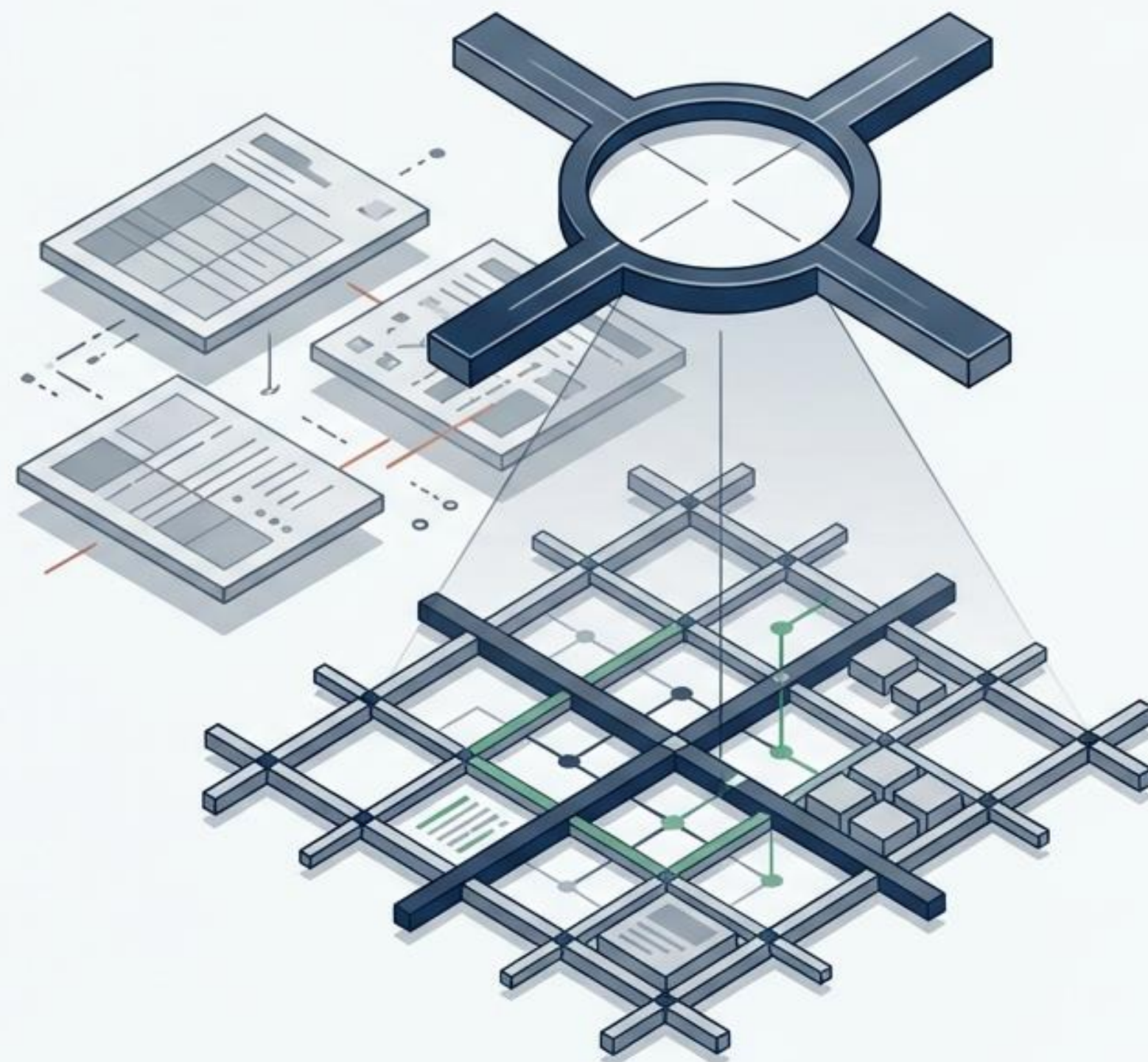
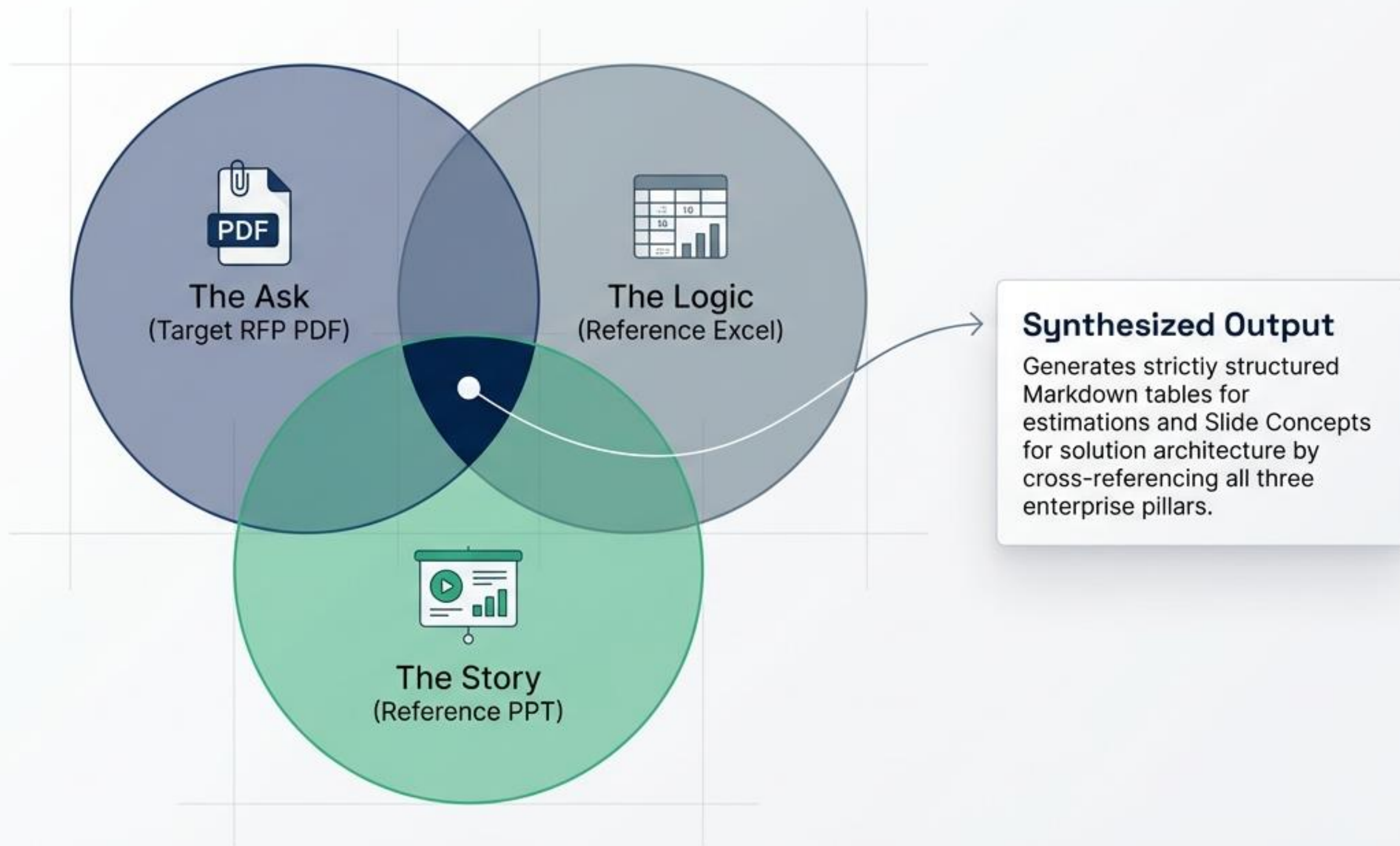


Reverse Engineering Winning Bids with Generative AI

- **Enterprise RFP Intelligence System:**
A purpose-built architecture replacing manual estimations.
- ⚡ **Core Objective:** Reverse engineer heuristics from past proposals to generate high-fidelity artifacts.
- **The Persona:** Operates under a strict Forensic Accountant logic for analytical rigor.



The Triangle of Truth Synthesizes Heterogeneous Data



A Modern, Scalable Foundation for Agentic AI



Frontend (UI)

- Next.js 15 (React 19)
- TypeScript & Tailwind CSS
- Node.js 20 Runtime



Backend (The Agent)

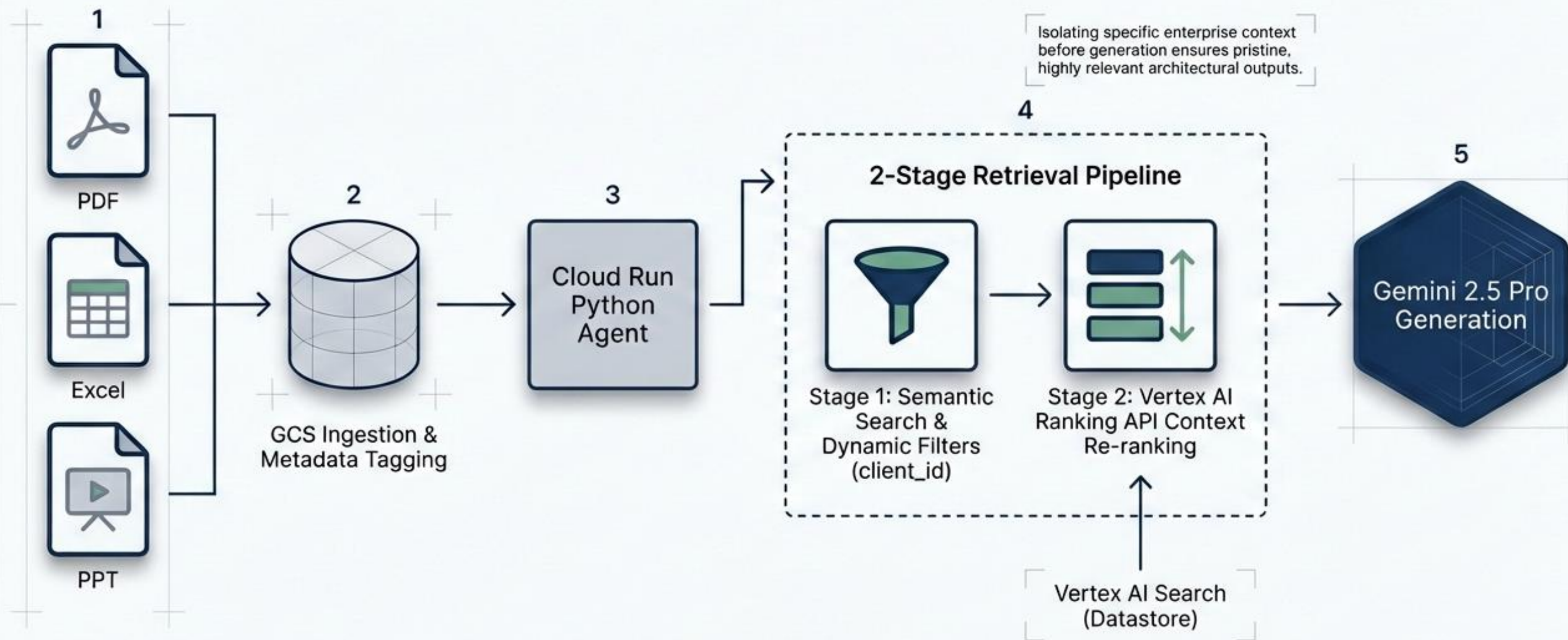
- Google Agent Development Kit
- Python 3.13
- gemini-3-pro-preview
- text-embedding-005



Infrastructure

- Google Cloud Run (Docker)
- Google Cloud Storage (GCS)
- Vertex AI Search (Datastore)

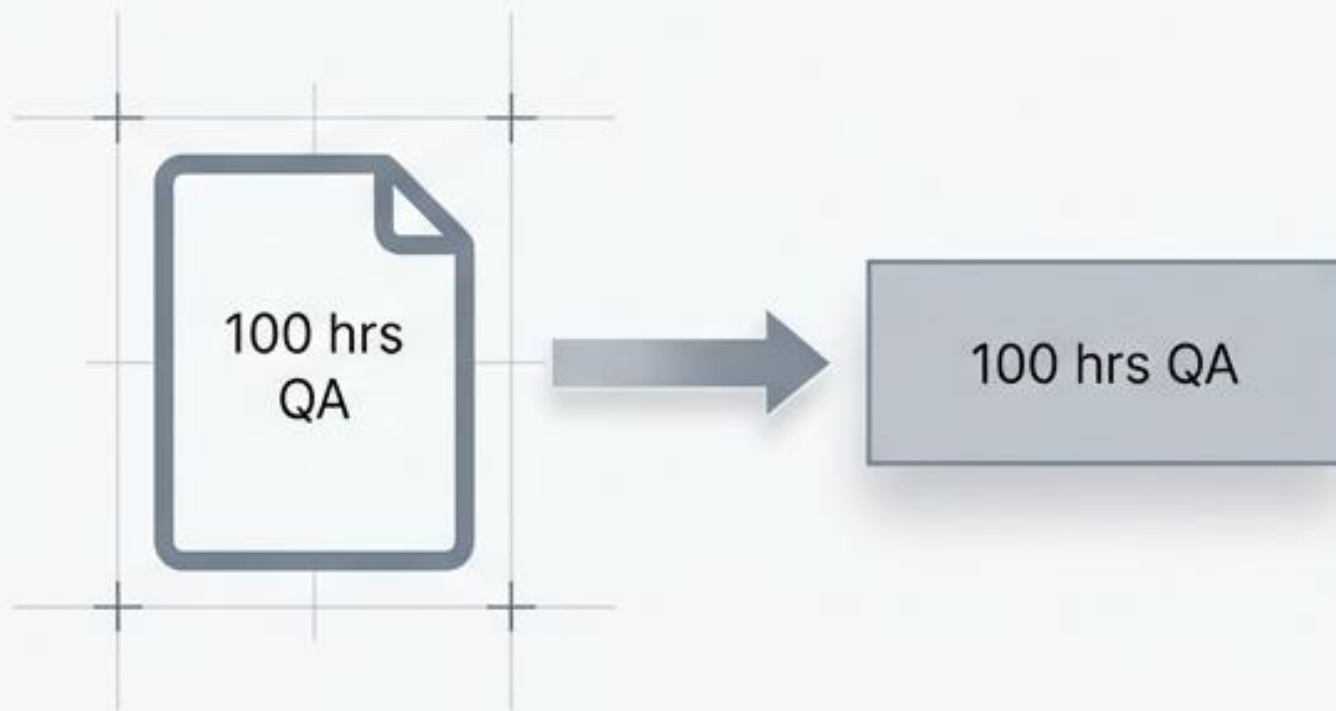
Orchestrating Multi-Modal RAG and 2-Stage Retrieval



Deriving Deep Heuristics Outperforms Shallow Data Extraction

Standard RAG Systems

Blindly extracts static numbers, failing on new scopes.



Forensic Accountant Logic

Actively derives underlying formulas to apply contextual math to new requirements.

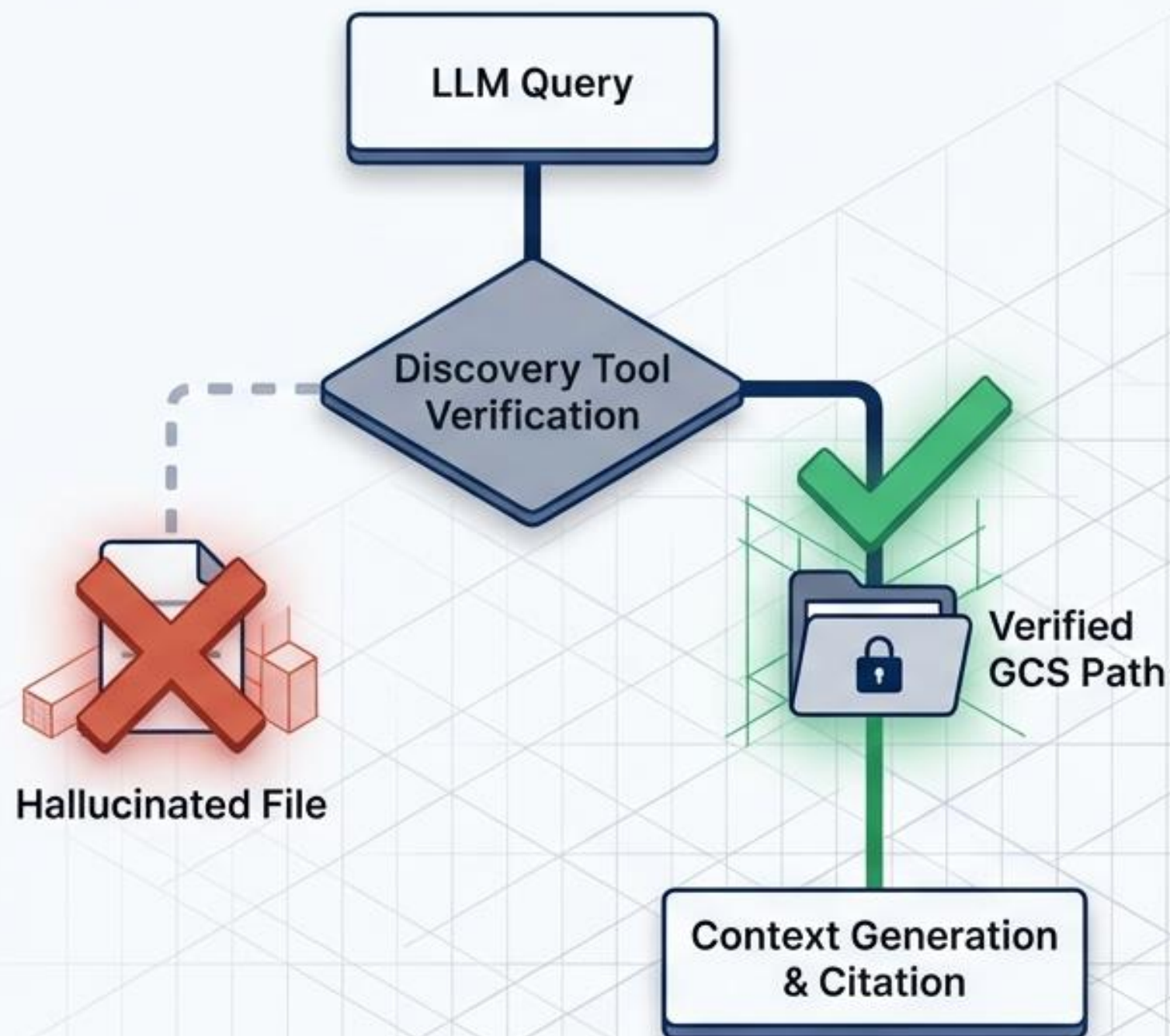


Strict Gatekeeping Eradicates Data Hallucinations

The Challenge: LLMs hallucinating non-existent files or fabricating estimation data.

The Discovery Tool: catalog.py physically executes `list_client_documents` in GCS prior to answering.

Precision Citations: Forced extraction of exact Excel Tab Names and Row Numbers.



Production-Ready Architecture Built for Enterprise Security



Automated RFP Impact

Delivers end-to-end response generation, drastically cutting manual estimation cycles.



Zero-Trust Security

Operates with strictly no hardcoded secrets, utilizing dynamic environment variables.



Enterprise Authentication

Secured natively via Google Application Default Credentials and IAM Workload Identity.



Serverless Scale

UI and Backend seamlessly containerized and deployed to Cloud Run for scale-to-zero efficiency.